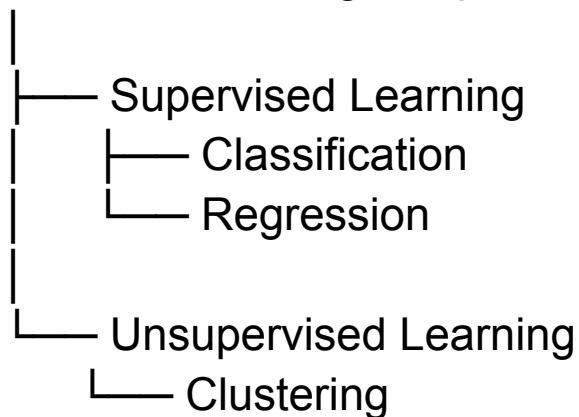


Classification, Regression, Clustering

Concept	One-Line Definition
Supervised Learning	Learning from labeled data.
Unsupervised Learning	Finding hidden patterns in unlabeled data.
Classification	Predicting categories or classes.
Regression	Predicting numerical values.
Clustering	Grouping similar data points together.

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Machine Learning/Deep Learning



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Dog / Cat / Plastic / Paper



Classification

₹500 / 35°C / 100 Marks



Regression

Group Similar Customers



Clustering

=====

Classification, Regression, Clustering

Technique	Description	Output Type	Example
Classification	Predicts which category or class an input belongs to.	Category / Label	Dog, Cat, Plastic, Paper
Regression	Predicts a continuous numerical value based on input data.	Number	₹500, Temperature 35°C, Marks 85
Clustering	Groups similar data points together without predefined labels.	Groups / Clusters	Customer Group A, Group B, Group C

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Quick Comparison Table

Technique	Main Goal	Output Example
Classification	Identify category	Dog, Cat, Plastic
Regression	Predict value	₹500, 35°C, 100 Marks
Clustering	Group similar data	Group A, Group B, Group C

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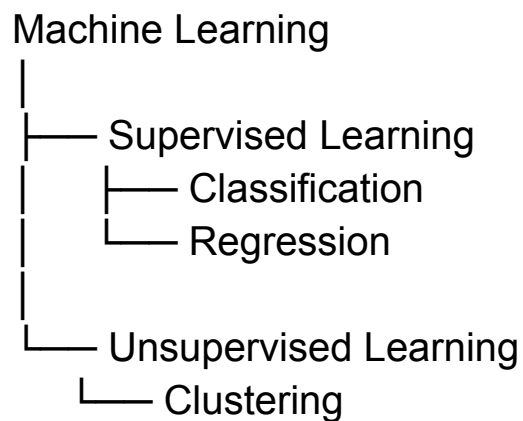
Problem	Technique	Learning Type
Dog Breed Identification	Classification	Supervised
Waste Classification	Classification	Supervised
Face Recognition	Classification	Supervised
House Price Prediction	Regression	Supervised

Stock Price Prediction	Regression	Supervised
Sales Forecasting	Regression	Supervised
Customer Segmentation	Clustering	Unsupervised
Tourist Segmentation	Clustering	Unsupervised
Market Basket Analysis	Clustering	Unsupervised

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Supervised/Unsupervised and **Classification/Regression/Clustering** are different concepts.

Think of it like this:



Relationship Table

Learning Type	Technique	Labels Available?	Example
Supervised	Classification	Yes	Dog vs Cat
Supervised	Regression	Yes	House Price Prediction
Unsupervised	Clustering	No	Customer Segmentation

Classification + Supervised

Dataset

Image	Label
Dog Image	Dog
Cat Image	Cat
Dog Image	Dog

The answer is already known.

Input Image → Dog

Input Image → Cat

Model learns from labels.

Example

Your Waste Classification Project

Input	Output
Bottle Image	Plastic
Newspaper Image	Paper

This is:

Supervised + Classification

Regression + Supervised

Dataset

House Size	Price
1000 sq.ft	₹20 Lakhs
1500 sq.ft	₹30 Lakhs

Labels already exist.

The model learns:

House Size → Price

Example

Stock Price Prediction

Today's Price	Tomorrow Price
₹100	₹105
₹105	₹110

This is:

Supervised + Regression

Clustering + Unsupervised

Dataset

Customer Age	Spend ing
22	High
23	High
55	Low
58	Low

No labels like:

Premium Customer

Normal Customer

The algorithm creates groups automatically.

Result

Cluster 1 → Young Customers

Cluster 2 → Senior Customers

This is:

Unsupervised + Clustering

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Learning Type	Description	Labels Available ?	Output Type	Sub Category	Example
Supervised Learning	Learns from data that already contains correct answers (labels).	Yes	Category or Number	Classification, Regression	Dog Breed Identification, House Price Prediction

Unsupervised Learning	Learns patterns from data without labels.	No	Groups / Hidden Patterns	Clustering, Association	Customer Segmentation, Market Basket Analysis
Reinforcement Learning	Learns by trial and error using rewards and penalties.	No	Actions / Decisions	Q-Learning, Deep RL	Self-Driving Cars, Game Playing AI

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Supervised Learning Subcategories

Technique	Description	Output Type	Example
Classification	Predicts which category an input belongs to.	Category / Label	Dog, Cat, Plastic, Paper
Regression	Predicts a continuous numerical value.	Number	₹500, 35°C, 85 Marks

Unsupervised Learning Subcategories

Technique	Description	Output Type	Example
Clustering	Groups similar data points together.	Groups / Clusters	Customer Group A, Group B, Group C
Association Rule Learning	Finds relationships between items.	Rules / Patterns	Customers buying Bread also buy Butter

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Classification

Description:

Determines the category of an object.

Example:

Input: Dog Image

Output: Labrador

Input: Waste Image

Output: Plastic

Used in:

- Dog Breed Identification
 - Waste Classification
 - Face Recognition
 - Spam Detection
-

Regression

Description:

Predicts a numerical value.

Example:

Input: House Size

Output: ₹50,00,000

Input: Study Hours

Output: 85 Marks

Used in:

- Stock Price Prediction
 - House Price Prediction
 - Sales Forecasting
 - Temperature Prediction
-

Clustering

Description:

Automatically groups similar data without knowing the answers beforehand.

Example:

Customer Data



Cluster 1 → High Spenders

Cluster 2 → Medium Spenders

Cluster 3 → Low Spenders

Used in:

- Customer Segmentation
- Tourist Group Analysis
- Market Research
- Recommendation Systems

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